

12. **Ans: C**

From Graph 1,  $\lambda = 0.45 - 0.15 = 0.30 \text{ m}$

From Graph 2,  $T = 1.6 - 1.2 = 0.4 \text{ ms} = 0.4 \times 10^{-3} \text{ s}$

$$v = f\lambda = \frac{\lambda}{T} = \frac{0.30}{0.4 \times 10^{-3}} = 750 \text{ m s}^{-1}$$

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